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Full length article

LR-STGCN: Detecting and mitigating low-rate DDoS attacks in SDN based on spatial–temporal graph neural network

Jin Wang, Liping Wang *

College of Computer Science and Technology, Zhejiang University of Technology, 310023, Hangzhou, China

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ABSTRACT

Software Defined Network (SDN) is an emerging network architecture. The decoupled data plane and control plane provide programmability for efficient network management. As a new network architecture, SDN also faces the threat of Low-rate Distributed Denial of Service (LDDoS) attacks. However, the centralized control, forwarding separation, scalability, and programmability of SDN provide new ideas for the detection and defense of LDDoS attacks. In this paper, we perform feature extraction of LDDoS attack flows in terms of time–frequency distribution of LDDoS attack flows and quality of service (QoS) of TCP flows, and identify the victim switch and victim ports by using the hybrid GCN-GRU deep learning model and the double sliding window method. Finally, the location of the attacking host is determined based on the victim port, and defense measures are issued to the victim switch at the attack source through the OpenFlow protocol. The evaluation results indicate that the detection method deployed on SDN controllers has a high detection rate and low false positive rate for LDDoS attacks, and can detect and alleviate LDDoS attacks online and in real-time.

1. Introduction

Distributed Denial of Service (DDoS) is one of the most common types of network attacks, posing a serious threat to the Internet. Low-rate Distributed Denial of Service attack (Kuzmanovic and E., 2003) is a special type of DDoS attack with periodicity and concealment. Unlike traditional DDoS attacks, attackers exploit design flaws in adaptive mechanisms (usually TCP Congestion Control mechanism) to periodically send high-capacity requests to attack targets through widely distributed puppet machines, triggering congestion repeatedly, significantly reducing the throughput of victim hosts and links, and affecting the transmission performance of the network. Due to the short pulse duration of each LDDoS attack, the average attack traffic is relatively small, which is very similar to the instantaneous burst traffic generated by many standard applications. As a result, LDDoS attack detection becomes very difficult.

In recent years, SDN (Xia et al., 2015) has become a development trend. As a new type of network innovation architecture, SDN achieves the separation of control plane and data plane compared to traditional network devices. The architecture of this new type of network is very obvious, simplifying the management and configuration of network equipment, forming a unified and efficient management and maintenance, improving the operational efficiency of the network, and reducing the maintenance cost of equipment. In addition, the openness and programmability of SDN enables network scalability. Developers can flexibly use high-level languages to design applications and modify network strategies on various layers of SDN without considering the underlying hardware. Similarly, the programmability of SDN provides new methods for detecting and mitigating LDDoS attacks. The attack detection algorithms in traditional networks can be directly deployed on SDN to achieve online detection. So far, there have been many studies on the detection and defense methods of SDN against other attacks, such as DoS/DDoS. However, there is little research on LDDoS attacks based on the TCP congestion control mechanism.

The advantage of studying LDDoS attacks in SDN is that SDN provides sufficient programmability. Based on SDN functionality and protocols, we can deploy LDDoS attack detection methods for real-time defense. However, in a highly centralized control-based SDN architecture, LDDoS attacks do not fundamentally change the congestion control mechanism, while also posing some unique threats to SDN. For example, LDDoS attacks target the hosts of the SDN network, attacking the core services on the control plane, thereby reducing the Quality of Service (QoS) of the entire SDN, and may even lead to network paralysis. Therefore, it is necessary to study LDDoS attacks in SDN.

In general, we face the following challenges in protecting SDN against LDDoS attacks on TCP congestion control:

- LDDoS attacks are highly concealed and similar to normal traffic. How to accurately detect LDDoS attack traffic in real-time.

* Corresponding author.
E-mail addresses: 1111912010@zjut.edu.cn (J. Wang), w59887711@gmail.com (L. Wang).

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- LDDoS attack traffic is sparse, scattered, and difficult to trace. How to effectively deploy the detection system and find the attack paths and sources in time.
- For LDDoS attacks in SDN, how to take effective measures to mitigate LDDoS attacks with low cost and acceptable time delay.

To address the above issues, this article proposes an LDDoS attack detection and mitigation system based on the OpenFlow protocol, i.e., LR-STGCN, and deploys the system into the SDN controller. The system consists of two parts: anomaly detection and mitigation. Firstly, in the anomaly detection stage, the time–frequency characteristics of the LDDoS attack flow and the QoS performance of TCP traffic before and after the attack are analyzed, and wavelet transform is used to extract features from the LDDoS attack flow. Then, the Graph Convolutional Network (Yang et al., 2022) (GCN) and Gate Recurrent Unit (Assis et al., 2021) (GRU) hybrid model GCN-GRU were used to detect anomalies in LDDoS attacks and identify switches with abnormal flows. In the mitigation stage, we use the double sliding window method to determine the location of the attack host. Then, the switch port connected to the attack host is blocked to prevent the attacker from sending LDDoS attack traffic again. The design of the scheme follows the OpenFlow protocol and does not require additional equipment. The main contributions of this paper are as follows.

- We propose a real-time online detection and mitigation system to defend against LDDoS attacks in SDN based on its centralized control, scalability, and programmability.
- To effectively extract the characteristics of LDDoS attacks, we analyze the time–frequency characteristics of LDDoS attack flows and the QoS performance of TCP traffic before and after the attack; And use wavelet transform to extract features from LDDoS attack flows, thereby improving the detection rate of LDDoS attacks.
- To effectively detect attacks, this paper adopts the GCN-GRU model. This model not only fully utilizes the spatial relationships between switches in network topology and the temporal characteristics of switch states to improve detection rate, but also detects the switches through which attack flows, thereby finding attack paths and attack hosts, and playing a beneficial role in tracing and evidencing attacks.

The rest of this paper is organized as follows. Section 2 provides a brief background knowledge of SDN architecture and LDoS attacks. Section 3 reviews the current research status of LDoS attacks in traditional networks and SDNs. Section 4 introduces the outline design of LR-STGCN. Section 5 provides a detailed description of LDDoS attack detection and mitigation solutions. Section 6 introduces the experimental design and results. Section 7 summarizes the entire text and explores future research directions.

2. Background

2.1. SDN and OpenFlow protocol

SDN is an emerging network architecture that mainly includes three planes and two interfaces, as shown in Fig. 1. The application plane includes various services and applications, such as intrusion detection systems and security monitors. The control plane is responsible for unified control of application plane programs and data plane forwarding devices through network devices, providing and adjusting performance services for the network. The data plane mainly includes forwarding devices such as routers and switches, which forward data packets based on controller management. The SDN control plane obtains underlying infrastructure information through a southbound interface and also provides an expandable northbound interface for the application plane.

OpenFlow is a communication protocol used to connect the control plane and the data plane in SDN, which is responsible for the interaction between the controller and the switch. In the OpenFlow protocol, the switch consists of a flow table and a secure channel, which are connected to the controller. The flow table consists of flow rules, each containing the matching field, the counter, and the operation. The match field is used to match incoming packets. The count field is used to count the packets matched by a certain flow rule. The action field defines the relevant actions that match the packet. In OpenFlow protocol, when a new packet arrives at the switch, the packet information is compared with the matching field of the flow rule, once the match is successful, the counter of the flow rule is updated, and the relevant operation is performed. If the packet information does not match the flow rule, the header information of the packet is encapsulated in the *packet_in* messages and sent to the controller for the corresponding flow rule request. Finally, the controller responds to the switch and sends the *packet_out* messages.

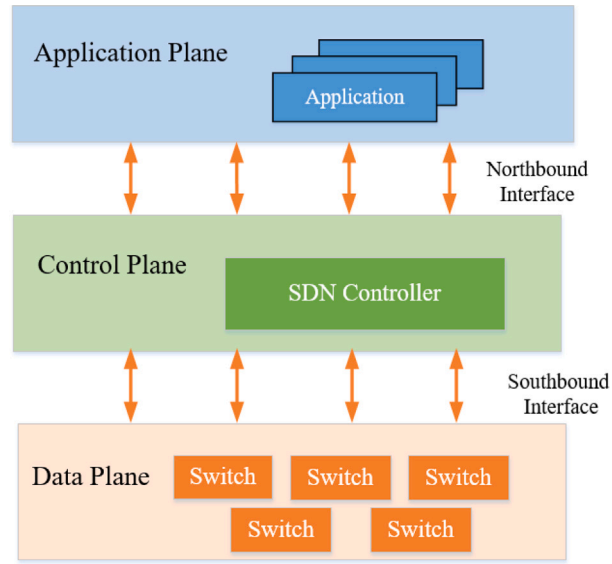


Fig. 1. SDN network structure.

2.2. LDoS attacks

LDoS attack is a new type of network attack in recent years, which mainly targets the vulnerabilities of various adaptive mechanisms in the network, such as TCP congestion control mechanism and router queue management mechanism. LDoS attacks use periodic high-rate, short-time pulses to make the network constantly change between the stable state and the unstable state, seriously reducing the performance of the network. There are two modes of attack, divided into the single-source LDoS attack and the distributed LDoS attack according to the number of attack sources.

(1) Single-source LDoS attack

The mathematical model of a single source LDoS attack pulse can be represented by a triple (R, L, T_0). Where R represents the attack rate; L represents the duration of a single attack pulse; T_0 stands for the attack period and represents the time interval between two consecutive attacks. The LDoS attack will send an attack pulse with attack duration L and attack rate R in each attack period T_0. When the attack intensity $(R \times L)$ is large enough, it can cause a sufficient number of packets to be lost, which triggers the RTO or AIMD mechanism to frequently adjust the number of packets to be sent to achieve the purpose of reducing the performance of the network. The theoretical model of a single-source LDoS attack is shown in Fig. 2.

(2) Distributed LDoS attack

Distributed LDoS attacks are coordinated by several attack sources, and there are two primary ways, as shown in Fig. 3.

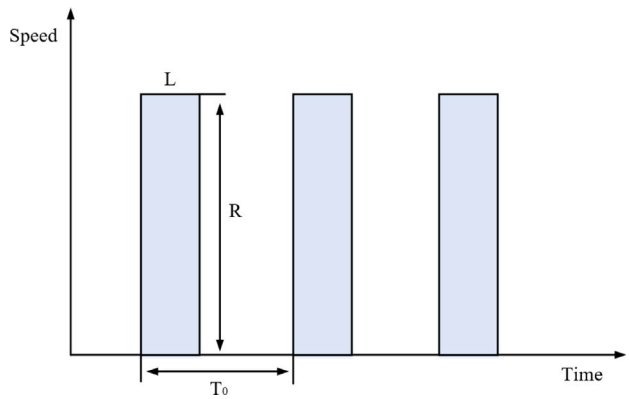


Fig. 2. Single-source LDoS attack model.

Fig. 3(a) is a synchronous distributed LDoS attack model, in which multiple attack sources (assuming n attack sources) simultaneously generate attack pulses with attack rate r ($r \geq R/n$) within the attack period T_0, and these attack pulses are clustered into high-intensity pulses (attack rate is about equal to R) at the victim side. This method requires high-time synchronization between attack sources and is difficult to realize. In this paper, we consider the synchronous distributed LDoS attack model.

Fig. 3(b) shows the asynchronous distributed LDoS attack model with multiple attack sources (assuming n attack sources), generating attack pulses with attack period $n \times T_0$ and attack rate R. These attack pulses form a complete pulse attack with attack period T_0 and attack rate R at the victim side. The asynchronous distributed LDoS attack does not require high-time synchronization, and the slight delay of the link has little effect on the attack effect.

2.3. TCP congestion control mechanism

The TCP protocol plays a vital role in the Internet. Most of the proposed LDoS attacks are aimed at various adaptive mechanisms in TCP, and the congestion control mechanism is the most important adaptive mechanism in TCP. Its main idea is that the sender automatically adjusts the packet sending rate according to the current link congestion. LDoS attacks against the TCP congestion control mechanism mainly use two adaptive mechanisms in the TCP congestion control algorithm: Retransmission Time Out (RTO) and Additive Increase Multiplicative Decrease (AIMD). The attack traffic with periodic short-time pulse characteristics causes normal TCP traffic to periodically drop packets, creating different degrees of network congestion, thus constantly triggering the RTO and AIMD of the two adaptive mechanisms, prompting the normal TCP sender's congestion control window has always been in a very small state, resulting in a sharp decrease in the victim host's TCP throughput, which seriously affects the performance of the network. This paper mainly considers distributed LDoS attacks with RTO and AIMD mechanisms, as shown in Fig. 4.

(1) LDDoS attack based on RTO mechanism

The RTO mechanism is a TCP congestion control mechanism based on the congestion window (cwnd). The TCP sender sets a timer for each packet sent. If the timer times out before receiving the ACK of the packet, the slow start algorithm is activated, and the RTO value is increased exponentially based on the exponential backoff algorithm. The sender's threshold (ssthresh) is set to half of the current cwnd, and the cwnd value is set to 1, and the packet is resent. If the attacker knows the RTO value, once the sender resends the packet, the attacker will generate a pulse attack, resulting in a timeout for each retransmission. The cwnd value is always set to 1, and the attack effect is shown in Fig. 4(a).

(2) LDDoS attack based on AIMD mechanism

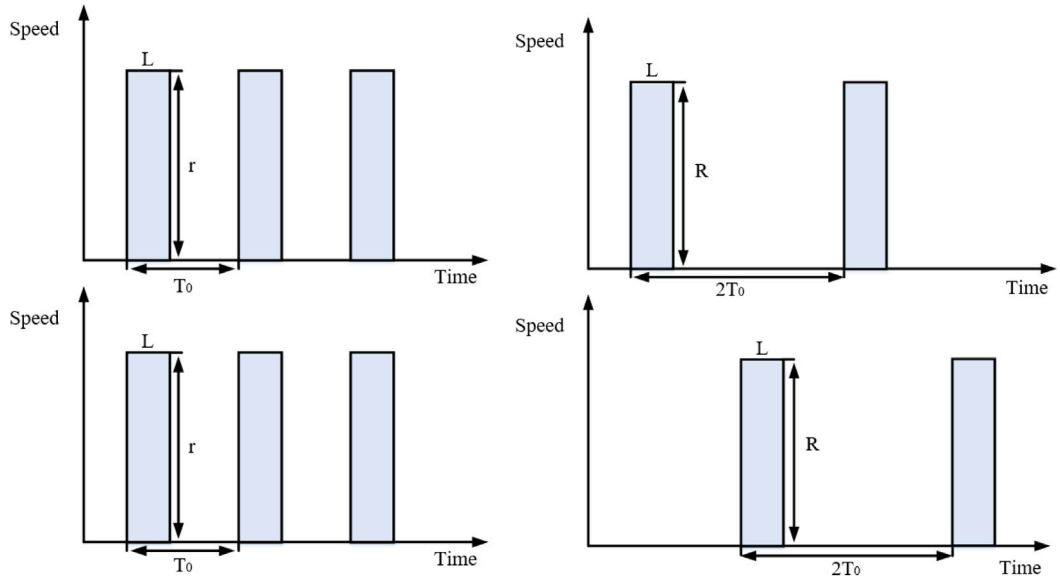
The AIMD mechanism differs from the RTO mechanism in that it adaptively adjusts the cwnd value when congestion occurs on the link. When the TCP sender receives three consecutive repeated ACK messages, the cwnd is reduced to half of its previous value and immediately resends this packet. As a result, the cwnd value continues to decline and eventually converges to a limit value, which greatly affects the TCP efficiency. Fig. 4(b) shows the attack effect.

3. Related work

Since Kuzmanovic first proposed to low rate denial of service (LDoS) attack, the research on detection and defense of LDoS attack has always been a concern in the field of network security. Traditional network LDoS attack detection methods are mainly considered from two aspects: frequency domain and time domain.

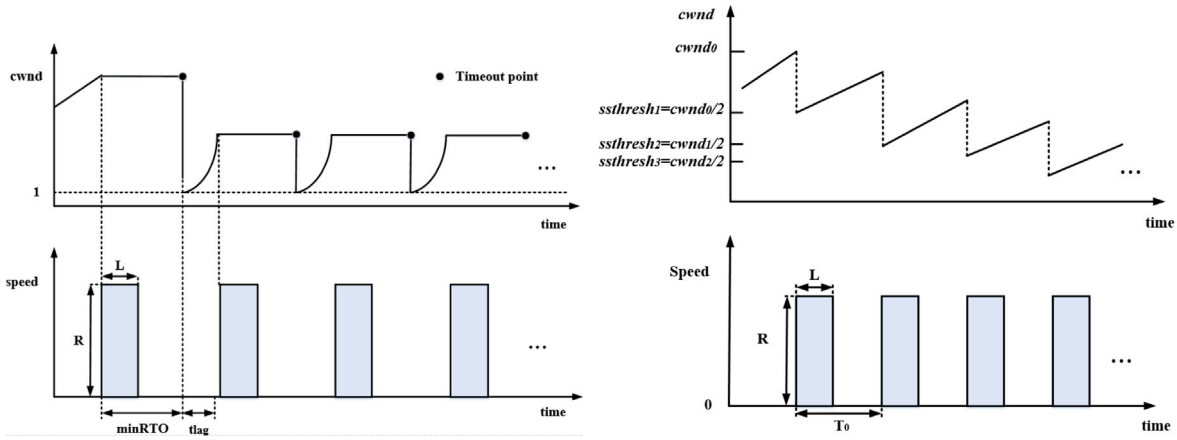
The frequency domain-based detection method mainly analyses the frequency of network traffic to determine whether it is similar to the frequency of LDoS attack traffic. The idea is to apply the classical signal detection theory and filtering theory to the detection method of LDoS attack traffic. Zhang et al. (2019) combined Support Vector Machine (SVM) with Power Spectral Density (PSD) based entropy to detect LDoS attacks. Wu et al. (2017) proposed an infinite Impulse Response Filtering (IIR)-based comb crash filter to filter out the attack streams in the frequency domain, which effectively filters out the LDoS attack streams while having less impact on legitimate TCP traffic. Chen et al. (2018) introduced Fourier energy spectral entropy and wavelet energy spectral entropy to detect LDoS attacks and proposed a queuing algorithm based on energy spectral entropy to mitigate LDoS attacks. Agrawal and S. (2020) used the fast Hartley transform to compute the PSD and detected LDoS attacks by analyzing the power spectral distributions. Agrawal and S. (2018) utilized the power spectral density method to detect LDoS attack traffic in cloud environments by analyzing the spectral distribution variability of legitimate traffic and LDoS attack traffic. Based on wavelet theory, Yue et al. (2018) proposed an LDoS attack traffic identification method based on the combination of wavelet transform and neural network according to the multiple fractal characteristics of network traffic. Zhan et al. (2020) transformed the TCP traffic in normal and attack states into the time–frequency domain using the short-time Fourier transform and obtained the joint time–frequency partitioning of the network traffic, then selected important statistical features based on the joint time–frequency distribution. Distribution to select the important statistical features for LDoS attack detection. Wu et al. (2018) used the Hilbert–Huang transform to obtain the Hilbert spectrum of the network traffic, obtain more information about the network traffic, and effectively detect LDoS attacks. Chen et al. (2019) used the HHT time–frequency joint distribution method for analyzing the non-smooth small signals generated by LDoS attacks. A combination of correlation coefficients and KS test methods are used to assess the confidence of the intrinsic modal function. The experimental results show that the method has high detection effectiveness. However, the frequency-domain based detection method is more accurate in extracting LDoS attack flow features. Still, the extraction process is more complicated and ignores the time-domain variation of the flow.

Time-domain based detection methods mainly analyze traffic sequences in the network and determine whether their characteristics match the traffic characteristics of LDoS attacks. Shi et al. (2022) proposed an LDoS attack detection algorithm combining a cloud model and SVM based on the distribution characteristics of TCP traffic under an LDoS attack. The cloud model morphology corresponding to TCP traffic under attack and normal state is quantitatively analyzed and numerical feature sets are extracted through cloud model theory, and then the SVM algorithm is borrowed to construct an anomaly detection model, which achieves the fast identification of LDoS attacks. Tang et al. (2020a) take advantage of the characteristics of the discrete nature of the legitimate traffic and the LDoS attack traffic to propose a Weighted Euclidean Distance (WEDMS)-based Advanced Mean Drift Algorithm,



(a) Synchronous distributed LDoS attack model (b) Asynchronous distributed LDoS attack model

Fig. 3. Distributed LDoS attack model for synchronous and asynchronous.



(a) RTO (b) AIMD

Fig. 4. LDoS attack based on RTO and AIMD mechanisms.

which clusters network traffic and finally determines the presence of LDoS attacks based on the decision features of the clustering results. Subsequently, they proposed a network traffic-based fractal residual real-time LDoS attack detection method based on the fractal residual characteristics of normal network and LDoS attack traffic (Tang et al., 2021a). Meanwhile, they also analyzed the legitimate traffic and LDoS attack traffic, extracted several network traffic time domain features (energy, occupancy, variance, entropy, etc.) as inputs for LDoS attack detection, and fused these features into a feature map by Convolutional Neural Networks (Samha et al., 2023) (CNN) model for LDoS attack classification (Tang et al., 2021b). In addition, they directly use these features as inputs to an improved Adaptive Boosting (AdaBoost) model to differentiate between LDoS attack traffic and normal traffic (Tang et al., 2020b). However, the time-domain based detection methods are simpler in feature extraction and are prone to high false alarm rates for periodic, short-duration high-speed pulses of LDoS attack flows.

With the rapid development of SDN architectures, some researchers have started to work on LDoS attack detection techniques for SDN environments. Agrawal and T. (2021) used the entropy values of the source IP address and packet size to detect LDDoS attacks and conducted experiments in real SDN-Cloud scenarios, and the results showed that the method was able to detect the LDDoS within 10 s attack and locate the source of the attack. Tang et al. (2023a) proposed a framework for LDoS attack detection and mitigation based on SDN architecture, i.e., GASF-IPP. The framework uses the Gram's Angle Field (GASF) transform based on temporal analysis to analyze the network traffic and perform the attack mitigation by downstream the flow rules. Experimental results demonstrate that the framework can detect LDoS attack flows in real time with a low false detection rate. Pérez-Díaz et al. (2020) used six machine learning models to train their proposed Intrusion Detection System (IDS), which is capable of detecting LDoS attack traffic effectively in real-time. Tang et al. (2022a) exploited the flexibility and scalability of SDN to propose an online attack detection and mitigation framework for LDoS attacks on SDN, named ADMS. The framework utilizes Lightgbm classifiers for anomaly detection of traffic features and flow table statistical features of switch ports, and the experimental results show that the detection module is able to accurately detect LDoS attacks. The mitigation module utilizes the Sequence Matching-based Dynamic Sequence Analysis Algorithm (SMDSA) to locate the attacker and effectively mitigate the LDoS attack flow through packet filters. Experimental results show that the ADMS system is able to accurately identify and effectively mitigate LDoS attacks in real time. Tang et al.

(2022b) proposed a lightweight real-time framework, $P\&F$, to detect and mitigate LDoS attacks on SDNs. The $P\&F$ framework identifies the LDoS attacks from both the attack performance and the characteristics of the attack flows, and according to these two aspects, proposes a joint detection method, which effectively defends against LDoS attack flows in SDNs. Tang et al. (2023b) proposed a novel framework for lightweight SDNs, PeakSAX, which protects SDNs from LDoS attacks in real time by continuously monitoring flow rules, identifying aggregated flows, identifying malicious flows, locating attackers, and deploying mitigation policies. Based on this, the programmability and centralized management model of SDN overcomes the shortcomings of off-line detection of LDoS attacks and achieves the advantages of online detection.

In summary, the current detection methods based on LDoS attacks mainly analyze the attack traffic from the port of a certain device, ignoring the dynamic and global characteristics of the traffic, which is prone to cause a high false alarm rate. Second, the current detection methods are only applicable to dumbbell attack topologies and lack research on complex multi-source attack topologies. Finally, there is a lack of effective real-time mitigation strategies for LDoS attacks against SDN. Therefore, how to improve the detection accuracy and efficiency of LDoS attacks urgently needs our solution.

4. System design

4.1. Network modeling

Due to the sparse and covert distribution of distributed low-rate attack flow, it is difficult to detect and alleviate it. To effectively detect the attack flow and identify the source of the attack for timely mitigation, we view the network topology as an undirected graph and solve the distributed low-rate attack flow detection problem by constructing a graph model.

The original collected network traffic date by time is divided according to the Detection Window (DW), which contains the traffic during a certain period by time. We assume the there are T consecutive DWs during the detection time. We denote the data collected by this network during T consecutive DWs is x_j^t for $j = 1, \dots, m$ and $t = 1, \dots, T$. Among them, x_i^t represents the data traffic collected by switch v_i $(i = 1, 2, \dots, m)$ on the network at tth DW. The purpose of this paper is to detect whether the data traffic x_i^t of switch v_i contains malicious traffic and identify the attack path in the network to determine the location of the attack source.

We establish a graph model $G = (V, E, A, X)$ to detect anomalies in switches, as shown in Fig. 5. V represents the set of nodes, where each switch represents a node. E represents the edge set, which represents the links between switches. A represents the adjacency matrix of graph G, X represents a three-dimensional tensor containing feature vectors of each node at different time periods, $X \in \mathbb{R}^{f \times m \times T}$ where f is the number of features of each node, m is the number of nodes, T is the number of DWs.

4.2. System overview

In SDN based on the OpenFlow protocol, the controller controls the forwarding behavior of the switch by issuing the flow table, and the flow table can be used to calculate the traffic information of each flow separately. Based on these characteristics of SDN, this paper proposes an LDDoS attack detection and mitigation system based on OpenFlow protocol, called LR-STGCN. LR-STGCN is deployed on the SDN controller and consists of an anomaly detection module and a mitigation module, and the LR-STGCN framework structure is shown in Fig. 6.

(1) Attack detection module

LDDoS attacks mimic normal user traffic behavior by operating at lower average rates and periodic pulses, causing their traffic patterns

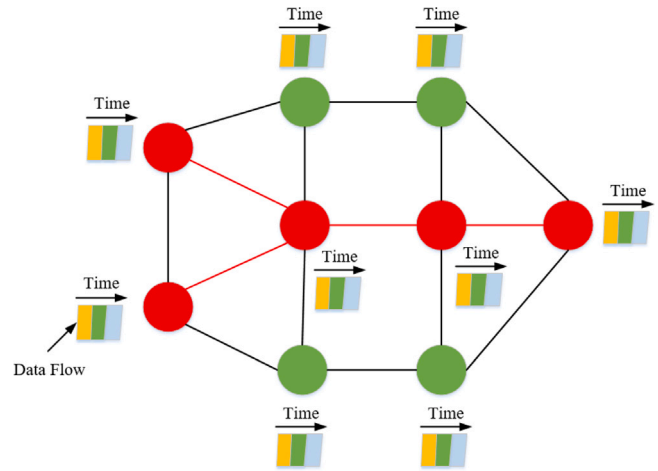


Fig. 5. Network model.

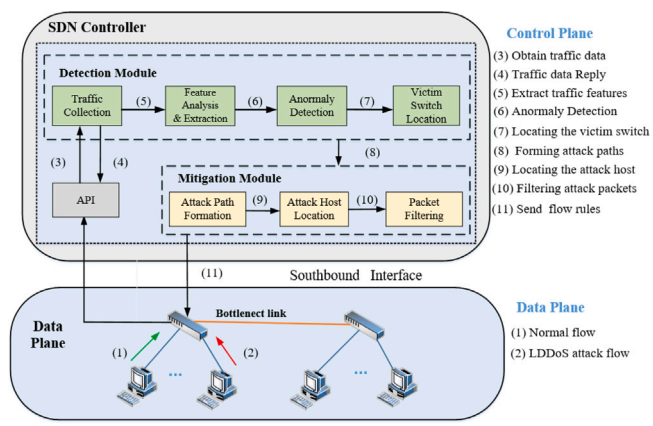


Fig. 6. Framework structure of LR-STGCN.

to closely resemble those of legitimate traffic. As a result, traditional detection methods based on rate, thresholds, or simple traffic features often fail to identify LDDoS attacks, making it difficult to distinguish attack traffic from legitimate traffic effectively. To better identify LDDoS attacks, this paper analyzes both the characteristics of LDDoS attacks and the impact of attack performance. On the one hand, by analyzing the characteristics of attack traffic. LDDoS attack has periodic and short-time high-speed pulse behavior characteristics, we can use the method of wavelet transform to extract the time–frequency characteristics of attack traffic. On the other hand, the performance impact of the LDDoS attack is analyzed. When such an attack occurs, the performance of TCP traffic degrades. We can extract the performance characteristics of the LDDoS attack (such as average bandwidth ratio, average rate, packet size, and so on) as the basis of anomaly judgment. Finally, the extracted features were input into the GCN-GRU model for fine-grained anomaly detection, to determine the switch through which the abnormal flow passed.

(2) Attack mitigation module

The purpose of the mitigation module is to effectively mitigate the LDDoS attack flow in SDN and ensure the normal operation of the network. The steps are as follows: First, the LR-STGCN system extracts the victim switch ID and constructs the attack path through the anomaly detection module. Then, the double sliding window algorithm and the attack path are used to locate the attack host. Finally, flow rules are issued to the victim switch ports at the attack source to filter out the attack packets.

5. Detailed design and implementation of the system

5.1. Detection module

5.1.1. Sampling process

To improve the attack detection rate, we need to accurately obtain various information in the network, such as statistics of ports, network flows, and data packets. In the traditional network, if you want to obtain this information, you need to deploy the corresponding data collection module in the network, but in the SDN environment, you only need to use OpenFlow technology to achieve this, and no additional collection equipment is needed. The controller obtains various statistical information in the switch through periodic polling and uses this information to calculate the real-time rate of each switch port or each flow.

The controller periodically sends Multipart Request messages to the switch to request the statistical information of the switch and calculates the rate of each port or data flow of the switch according to the statistical information returned by the switch. The total reception rate of packets at the switch port is calculated as follows.

$$X_i(n) = \frac{b_n - b_{n-1}}{t_n - t_{n-1}} \tag{1}$$

Among them, b_n is the number of packets received by switch v_i in the flow_states_reply messages during the nth round of polling. When t_n is the nth round of polling, calculate the time based on the duration_sec or duration_nsec field values in the flow_stats_reply messages, where duration_sec and duration_nsec represent the time when the current flow table already exists. Then $X_i(n)$ is the average rate of packets received by switch v_i at the nth poll. Similarly, the flow rate can be obtained using similar methods, providing a basis for the next step of feature extraction.

5.1.2. Feature extraction

Due to the frequent presence of a large amount of noise in network traffic, LDDoS attack flows are easily overwhelmed by noise. Analyzing LDDoS attack flows directly poses certain challenges. Therefore, it is necessary to find critical information from LDDoS attack flows. This paper calculates the statistical values of flow in the Detection Window (DW) and uses them as statistical features for LDDoS attacks. These features are mainly considered from two aspects: the time–frequency and attack performance of the attack flow.

Due to the fluctuation of LDDoS attack flows, as well as its strong periodicity and impulsivity. Therefore, this paper extracts features from the frequency domain perspective for LDDoS attack flows. Wavelet transform is a signal analysis tool with a fixed time–frequency area and adjustable shape, which has unique advantages in time–frequency localization and is very suitable for signal anomaly detection (Burrus et al.). The idea of time–frequency feature extraction based on attack flow is: firstly, the binary discrete wavelet transform is used to decompose the network traffic packet into J-layers, as shown in Eq. (2), $\{X(t), t = n\Delta t, n = 1, 2, \dots, N\}$ is a packets process, where $X(t)$ is number of data packets passing through the switch during the time $(t - \Delta t, t]$, and Δt is the sampling interval, and N is the number of samples in a DW.

$$X(t) = \sum_{j=1}^{J} \sum_{k} d_{j,k}\psi_{j,k}(t) + \sum_{k} a_{J,k}\varphi_{J,k}(t) \tag{2}$$

Among them, $\{\psi_{j,k}(t) = 2^{-j/2}\psi(2^{-j}t - k), j, k \in \mathbb{Z}\}$ are wavelet functions, $\{\varphi_{J,k}(t) = 2^{J/2}\varphi(2^{-J}t - k), k \in \mathbb{Z}\}$ are the scaling functions at scale J, $d_{j,k}$ are the wavelet coefficients of $X(t)$, $a_{J,k}$ are the scaling coefficients of $X(t)$ at scale J. Then, due to the differences in frequency domain distribution between LDDoS attack flows and normal flows, specific scale wavelet coefficients, and scaling coefficients are selected to reconstruct signals to distinguish LDDoS attack flows. This paper uses the Daubechies wavelet Db4 to perform the 5-layer wavelet decomposition (i.e., $J = 5$) on the packet process (S.G., 1989). Denote the low-pass filters and high-pass filters by $h_0, h_1, \dots, h_7$ and $g_{-6}, g_{-5}, \dots, g_1$ of Db4 respectively, where $g_k = (-1)^k h_{1-k}$. Then we get the decomposition formulas

$$a_{j,k} = \sum_{n=2k}^{2k+7} h_{n-2k} a_{j-1,k} \tag{3}$$

$$d_{j,k} = \sum_{n=2k-6}^{2k+1} g_{n-2k} a_{j-1,k} \tag{4}$$

and the reconstruction formulas

$$a_{j-1,n} = \sum_{k=\lceil (n-7)/2 \rceil}^{\lfloor n/2 \rfloor} h_{n-2k} a_{j,k} + \sum_{k=\lceil n/2 \rceil}^{\lfloor (n+7)/2 \rfloor} g_{n-2k} d_{j,k} \tag{5}$$

where $\lfloor x \rfloor$ represents the maximum integer not exceeding x, and $\lceil x \rceil$ represents the smallest integer not less than x.

Take $a_{0,k} = X(k\Delta t)$, $k = 1, 2, \dots, N$, by the decomposition formulas, we obtain 5 layers wavelet decomposition. By the reconstruction formulas, the 3, 4, and 5 layers of wavelet coefficients ($d_{3,k}, d_{4,k}, d_{5,k}$ and $d_{1,k} = d_{2,k} = a_{5,k} = 0$) are selected to reconstruct the attack flow sub-band $X_a(n)$ (He et al., 2009) that reflects the low-frequency band change information. Based on the analysis of LDDoS attack features, four LDDoS attack features with low computational complexity and high discrimination are extracted from the wavelet coefficients and $X_a(n)$, and this feature set was marked as A, as described below:

(1) Time–frequency features of attack flow

(1) The average value Avg_a of the attack flow sub-band $X_a(n)$.

The low-rate attack flow has a long cycle and is concentrated in the low-frequency band of the wavelet decomposition of the network flow. In the wavelet decomposition of normal network flow, it is concentrated in the high-frequency band, so the average value of Avg_a can distinguish low-speed attacks. The attack flow sub-band reflects the overall trend of background traffic of legitimate users. Under normal circumstances, the legitimate user data traffic basically remains stable, the effective utilization of network bandwidth is very high, and the average network traffic. The average data traffic of legitimate users decreases significantly, and Avg_a decreases.

$$Avg_a = \frac{\sum_{n=1}^{N} |X_a(n)|}{N} \tag{6}$$

(2) The standard deviation S_a of the attack flow sub-band $|X_a(n)|$.

LDDoS is a periodic pulse-type attack, with unstable network traffic and significant volatility. Its volatility is reflected by the standard deviation S_a of attack sub-band $|X_a(n)|$. When the attack is active, network traffic fluctuates significantly and increases; When the attack enters the sleep period, the data flow is transmitted at a smaller rate, and the fluctuation of network traffic is not significant, resulting in a very small $|X_a(n)|$.

$$S_a = \sqrt{\frac{\sum_{n=1}^{N} (|X_a(n)| - E(|X_a|))^2}{N-1}} \tag{7}$$

Among them, $|X_a(n)|$ represents the strength of the LDDoS attack flow, and $E(|X_a|)$ represents the average intensity of the LDDoS attack flow.

$$E(|X_a|) = \sum_{n=1}^{N} |X_a(n)|/N \tag{8}$$

(3) Attack frequency band energy E_a.

Based on these wavelet coefficients, we can define the wavelet energy of signal $X(t)$ at scale j as:

$$E_j = \frac{1}{D_j} \sum_{k} |d_j(j,k)|^2 \tag{9}$$

where D_j is the number of wavelet coefficients at the corresponding decomposition level. Therefore, E_j represents the wavelet energy for the frequency band ranging from $f/2^{j-1}$ to $f/2^j$, $[0, f]$ is the frequency ranges of $X(t)$.

E_a is the sum of energy at various scales within the attack frequency band. For normal network data, most of it is TCP traffic, concentrated in the high-frequency band, with low low-frequency energy and low E_a. When the periodic LDDoS attack is launched, the legal background data of high frequency is suppressed, and the energy is concentrated in the low-frequency band so that the low-frequency energy E_a becomes larger.

$$E_a = \sum_{j \in Attackband} E_j \tag{10}$$

(4) Wavelet Energy Entropy $WPSE$.

From the perspective of energy change, wavelet energy spectrum entropy can reflect the energy spectrum distribution of different network traffic. Therefore, the existence of LDDoS attacks can be well characterized by wavelet spectrum entropy. The low entropy of the wavelet energy spectrum indicates that the energy spectrum distribution of network traffic in each frequency band is relatively dispersed. The high entropy of the wavelet energy spectrum means that the energy spectrum distribution of network traffic in each frequency band is relatively concentrated. In the normal network, the energy spectrum distribution is relatively concentrated, and the entropy of the wavelet spectrum is large. When the network is attacked by LDDoS, the high-frequency component decreases due to the suppression of the attack, and the energy of the low-frequency part increases, so the corresponding wavelet energy value will be relatively dispersed and the wavelet energy spectrum value will be small. Wavelet spectrum entropy $WPSE$ is defined as:

$$WPSE = -\sum_{k=1}^{n} p_k \log p_k \tag{11}$$

Among them, $p_k = E_k/E$, and $E = \sum_{j=1}^{n} E_j$ be the total power of the signal.

(2) Attack performance features

In order to improve the detection accuracy of the model and ensure the reliability of the results, this paper analyzes the traffic characteristics from the aspect of TCP traffic service quality degradation, and the traffic features set is denoted as B. The B set includes the following four statistical features, which are described as follows:

(1) TCP bandwidth ratio $Ratio_{band}$.

The LDDoS attack is a type of degraded service attack. Once attacked, according to the TCP adaptive protocol, the TCP bandwidth immediately decreases and the TCP bandwidth ratio also decreases.

$$Ratio_{Band} = \frac{\sum flow_i}{N\Delta t \times bw} \tag{12}$$

Among them, $flow_i$ represents the number of TCP packets passing over the link bandwidth within the detection window, and bw represents the link bandwidth size.

(2) The correlation coefficient $Coeff_{pkt}$ of data packets.

During the LDDoS attack, due to the periodic triggering of attacks by attackers, the fluctuation of TCP traffic will be greater than normal network traffic. Therefore, we use variables related to data packets to reflect the fluctuation of TCP traffic.

$$Coeff_{pkt} = \frac{std(pkt)}{avg(pkt)} \tag{13}$$

Among them, $std(pkt)$ represents the variance of TCP packets in the detection window; $avg(pkt)$ represents the average number of TCP packets in the detection window.

(3) Average packet transmission speed Avg_{speed}

Due to LDDoS attacks, network service quality decreases, and the system initiates congestion avoidance algorithms, reducing TCP network bandwidth and TCP traffic, resulting in a decrease in TCP packet transmission rate.

$$Avg_{speed} = \frac{\sum_{i=1}^{num} pkt_i}{N\Delta t} \tag{14}$$

Among them, $\sum_{i=1}^{num} pkt_i$ represents the total number of TCP packets in the detection window.

(4) Average packet size Avg_{size}.

LDDoS attacks usually use a large number of smaller size packets to fill buffers in network devices, resulting in the loss of normal TCP packets. Therefore, the packet size becomes smaller when the LDDoS attack occurs.

$$Avg_{size} = \frac{\sum_{i=1}^{n} packet_size_i}{n} \tag{15}$$

Among them, n represents the number of packets within the detection window; $packet_size_i$ represents the size of each packet.

We fuse the two types of LDDoS attack features of sets A and B together and input the collected statistical features into the GCN-GRU anomaly detection model for LDDoS attack judgment.

5.1.3. GCN-GRU based intrusion detection model

The LDDoS attacks follow the form of cycles and pulses in time. In addition, when LDDoS attack flows enter the SDN network and forward between switches, it will have different impacts on the state of different switches, so the differences in switch states must follow certain rules in space. Considering these factors, we attempt to model and analyze switch status from both a temporal and spatial perspective when the LDDoS attack occurs, identify the switches affected by the attack, and then implement effective mitigation measures for the affected switches. Based on this, this article proposes a detection method for the control plane programmable SDN based on the spatial–temporal graph convolutional network (GCN-GRU). This detection model maps the SDN network into a graph, where each switch is treated as a node, and the connection relationships between switches are treated as edges. We collect flow information from each switch port through wireshark, input network flow information into the GCN-GRU detection model, and ultimately identify the switches affected by the LDDoS attack. The structure of the GCN-GRU model is shown in Fig. 7. The main structure of this model is divided into three parts, namely the input layer, spatial–temporal processing layer, and output layer. The input layer receives the switch flow data X and the adjacency matrix A of its graph model and uses it as input for the spatial–temporal processing layer. The spatial–temporal processing layer consists of GCN modules and GRU modules. Firstly, the GCN module receives data from the input layer and extracts the spatial features of the switch data through graph convolution operations. Then, it will be used as input to the GRU module. Finally, the GRU module is used to extract the temporal features of the switch data and transmit them to the fully connected layer. The output layer obtains classification results through a fully connected layer combined with the softmax function. The output layer results are divided into two categories: switch normal and abnormal. The GCN-GRU anomaly detection model algorithm is shown in Algorithm 1.

(1) GCN module based on spatial feature extraction

In the LDDoS attack, the aggregation of attack traffic on congested links is necessary to trigger the attack implementation. Therefore, the traffic characteristics of affected switches are interrelated, and using graph neural networks to extract node representation features is beneficial.

Graph Convolutional Neural Network (GCN) is a deep learning model that focuses on graph structures. GCN can extract spatial features from data through graph convolution operations and is commonly used to handle prediction and classification problems on non-Euclidean spatial data. For the graph $G_t = (V, E, A, X)$, where A represents the adjacency matrix of the graph G, X is the feature matrix of the node of G. According to the theory proposed by Kipf and M. (2017), The graph convolution operation of one-layer is defined as Eq. (16):

$$Z = \sigma(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} XW) \tag{16}$$

In Eq. (16), $\tilde{A} = A + I$ represents the sum of the adjacency matrix A and the identity matrix I, where I is the identity matrix of size $m \times m$; $\tilde{D}$ represents the diagonal matrix of $\tilde{A}$, W is a parameter matrix,

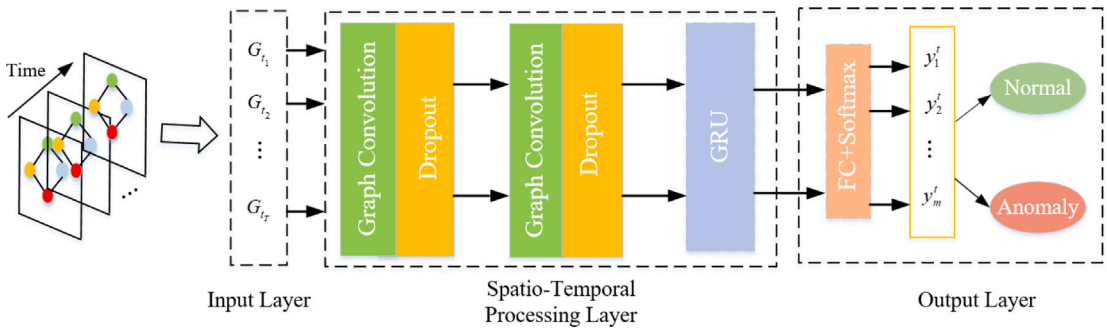


Fig. 7. GCN-GRU Structure.

$\sigma(\cdot)$ denotes an activation function. The one-layer GCN can capture information the direct neighbors. Let $\hat{A} = \tilde{D}^{-\frac{1}{2}}\tilde{A}\tilde{D}^{-\frac{1}{2}}$. To incorporate higher-order neighborhoods information, it would stack several GCN layers:

$$Z^{(l+1)} = \sigma(\hat{A}Z^l W^l) \tag{17}$$

where l is the layer number and $Z^{(l+1)}$ represents the output value of the lth graph convolution layer, σ represents the nonlinear activation function; W^l represents the parameter matrix of the lth graph convolution layer. Based on the article by Kipf and M. (2017), $l = 2$, i.e. the two-layers of GCN is suitable.

We consider network Graphs $G_{t_1}, G_{t_2}, \dots, G_{t_T}$ of T DWs, see Fig. 6, input switch data X_t and the adjacency matrix A of the graph, extract the spatial feature s^t of the data through graph convolution operation at each time t_i for $i = 1, 2, \dots, T$. We take the 2-layer graph convolution formula represents as follows:

$$s^t = \sigma(\hat{A}\sigma(\hat{A}X_t W^1)W^2) \tag{18}$$

Where $t = 1, 2, \dots, T$. As mentioned in Section 5, we let $u_{(v_i,t)}$ denote the feature vector of node v_i $(i = 1, 2, \dots, m)$ at the tth DW. We construct a two-dimensional span $X_t = (u_{(v_1,t)}, u_{(v_2,t)}, \dots, u_{(v_m,t)})$ $(t = 1, 2, \dots, T)$ as the inputs of the graph convolution network.

(2) GRU module based on temporal feature extraction

Network traffic is a Markov process, and the features of the inspection window series are correlated in time. Extracting the sequence features of the inspection window is beneficial for reflecting the characteristics of network attack traffic.

Another key to network traffic detection is to obtain the temporal correlation between the data. Traditional Recurrent Neural Network (RNN) has limitations for long-term prediction, and both Long Short-Term Memory Network (LSTM) and Gated Recurrent Unit (GRU) are variants of RNN, which can overcome the shortcomings of RNN. However, the structure of LSTM is complex and the training time is long. However, GRU has a relatively simple structure, fewer parameters, and faster training speed. Therefore, this paper chooses the GRU module to extract the time characteristics of network traffic of the nodes.

$$h^t = F_{GRU}(x^t, h^{t-1}) \tag{19}$$

The explicit formula as follows:

$$z_t = \sigma(W_z x^t + U_z h^{t-1} + b_z) \tag{20}$$

$$r_t = \sigma(W_r x^t + U_r h^{t-1} + b_r) \tag{21}$$

$$\tilde{h}^t = tanh(W_h x^t + U_h(r_t \odot h^{t-1}) + b) \tag{22}$$

$$h^t = (1 - z_t) \odot h^{t-1} + z_t \odot \tilde{h}^t \tag{23}$$

Among them, r_t represents the reset gate, z_t represents the update gate, h_{t-1} represents the output of the previous moment and s_i^t represents the input of the current time. W_z, U_z, W_r, U_r, W_h and U_h represent the weight matrix; σ represents the sigmoid activation function; $\odot$ represents the Hadamard product. Where the update gate z_t is used to determine the degree of influence of the output h^{t-1} at the previous time on the output h^t at the current time. GRU receives the spatial feature s_i^t $(t = 1, \dots, T)$ of each node v_i with T DWs extracted by the GCN module and uses it as input of f_{GRU} to obtain the temporal feature H^t of graph G_t at tth DW:

$$h_i^t = f_{GRU}(s_i^t, h_i^{t-1}) \tag{24}$$

Where $s_i = (s_i^1, s_i^2, \dots, s_i^T)$ represents the feature of node v_i from the Eq. (18) for $i = 1, 2, \dots, m$. Denote the output features of the GRU module by h_i^t which represents the new feature of node v_i for the T DWs from Eq. (24). Finally, from the above two module we obtain the spatio-temporal features H^t of nodes of graph G_t at tth DW for $t = 1, \dots, T$, then put these features as the input of the fully connected layer to obtain the classification probability Y^t of nodes in a graph at tth DW. The forward propagation formula of the fully connected layer in the GCN-GRU model is shown in Eq. (25).

$$Y^t = \sigma(W_f H^t + b) \tag{25}$$

Where W_f represents the weight matrix of the fully connected layer; b represents bias; H^t is the features of nodes in the graph at tth DW; $Y^t = [y_1, y_2, \dots, y_m]^t$ represents the classification probability matrix, $y_i = [1, 0](i = 1, 2, \dots, m)$ represents the classification probability of v_i, $y_i = 0$ represents the normal switch, $y_i = 1$ represents the malicious switch.

Algorithm 1 LDDoS attack detection model based on GCN-GRU

Input: Flow data X, Adjacency matrix A, Weight matrix W^0, W^1, W_r, W_z, W_h, Switch v_i,
Output: Classification results of each switch *result*.

```
Ã = A + I
D̃ = Σ_{j=1}^{n} Ã_ij
Â = D̃^{-1/2} Ã D̃^{-1/2}
for i = 1, 2, ..., m do
    for t = 1, 2, ..., T do
        s^t = f_GCN(X_t, Â) = σ(Â(σ(Â X_t W^0) W^1))
    end for
    Obtain spatial feature set S: S ← [s_i^1, s_i^2, ..., s_i^T]
    for t = 1, 2, ..., T do
        h_i^t = f_GRU(s_i^t, h_i^{t-1})
    end for
    Obtain the spatio-temporal feature set H of switch v_i at each DW: H ← [h_i^1, h_i^2, ..., h_i^T]
    Calculate each switch classification probability of v_1, v_2, ..., v_m at t-th DW: Y^t = σ(W_f H^t + b)
end for
result ← [y_1, y_2, ..., y_m]
```

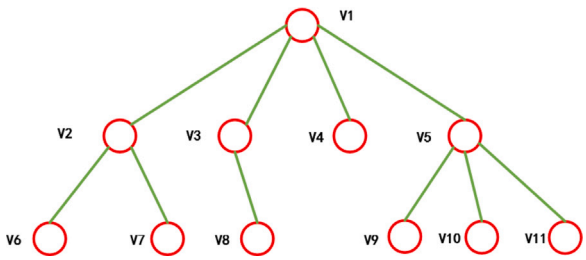


Fig. 8. Attack path tree.

5.2. Mitigation module

After detecting the LDDoS attack, the LR-STGCN system activates the mitigation module. This module consists of three components: (A) Locate the victim switch with the attack source; (B) Locate the victim port; (C) Deployment of defense strategies. The main idea is: First, determine which victim switches are connected to the attacking host by calculating the attack band energy E_a for each victim switch. Second, identify the victim ports of the switches (i.e., the ports connecting the switches to the host) using the double sliding window method. Finally, locate the attacking host and deploy the defense strategy. The specific description is as follows:

(A) Locate the victim switch with the attack source

Through the detection module, the switch ID through which the malicious flow passes is determined. The attack path is constructed into an attack tree graph according to the victim switch ID and network topology, as shown in Fig. 8. Among them, nodes are victim switches, edges are the connection relationship of victim switches, leaf nodes $(v_6, v_7, \dots, v_{11})$ are victim switches connected to the attack source, root (v_1) is the victim switch connected to the bottleneck link, and intermediate nodes (v_2, v_3, v_4, v_5) are divides. Next the attack host at the intermediate node is determined. Take $E_{min} = \min\{E^i_{attack}\}$, where E^i_a is the energy of victim switch v_i. Then $R^j_a = E^j_a / E_{min}$ is almost positive integer for every victim switch v_j. We define the vertex weight by $w_j = \sqrt{R^j_a}$ for vertex v_j. The root of the tree is v_k, where $k = \arg\max_i\{E^i_a\}$. Suppose that v_j has children $v_{j_1}, \dots, v_{j_k}$. According to the superposition property of wavelet transform, if $w_j > w_{j_1} + \cdots + w_{j_k}$ in the case of synchronous distributed LDoS attack, or $E^j_a > E^{j_1}_a + \cdots + E^{j_k}_a$ in the case of asynchronous distributed LDoS attack with $N/2^J \geq N\Delta t/T_0$, then switch v_j has least one attack source.

(B) Locate the victim port

Due to the impulsive nature of LDDoS attacks, the double sliding window method is used to find out the switch port that the malicious flow passes through, as shown in Algorithm 2, so as to determine the location of the attack host. The steps are as follows:

Step1: Set relevant parameters. Set the sampling time to t', and the length of the detection slices DS_A and detection slices DS_B to N. The setting of N_0 value is calculated based on the attack pulse length L. If the sampling time interval is τ, the sliding window N_0 is L/τ.

Step2: In the detection slice DS, if the network flow rate at switch port i is $g_i(n)$ during the nth polling, the energy values in DS_A and DS_B are calculated as:

$$E_A(n) = \sum_{k=0}^{N_0-1} g_i(n+k) \tag{26}$$

$$E_B(n) = \sum_{k=N_0}^{2N_0-1} g_i(n+k) \tag{27}$$

Step3: Calculate the energy ratio R of network flow in DS_A and DS_B:

$$R = \frac{E_B}{E_A} \tag{28}$$

Table 1
Defense rules.

Match field	Priority	Action
'In_port': Attacked port 'Ip_proto': UDP	Very high	Drop

Step4: If R is greater than the set threshold λ, it indicates that there are multiple consecutive large sampling values, and it can be considered that an abnormal pulse has occurred at this time.

Step5: If the number of abnormal pulses exceeds the set threshold K in the detection window, it can be determined that attack flows pass through this port and this port is defined as the victim port. The host connected to the victim port is the attacking host. The specific description is shown in Algorithm 3.

Algorithm 2 Victim port location

Input: Switch $vswitch$, Switch port flow rate $g(x)$, Detection slice length N_0, thresholds λ, K.
Output: Victim port $VictimPort$.

```
function VictimPortLoc(vswitch, g(x), N0, λ, K)
    for i = 1; i <= vswitch.port.length; i++ do
        Calculate E_A and E_B by the Eqs.(26) and (27), Calculate R by the Eq.(28)
        if R ≥ λ then
            Burst_Signals++
        end if
        if Burst_Signals ≥ K then
            VictimPort[k++] ← vswitch.port
        end if
    end for
end function
```

Algorithm 3 Attack host location

Input: $AttackTree$, $VictimSwitch$.
Output: $AttackerIP$.

```
for j = 1; j <= VictimSwitch.length; j++ do
    if VictimSwitch[j] is a inner node & w_j > w_{j_1} + ··· + w_{j_k} then
        VictimPort[k] ← VictimPortLoc(VictimSwitch[j])
        if VictimPort[k] is connected to a host then
            AttackerIP ← host.IP
        end if
    elseVictimSwitch[j] is a leaf node
        VictimPort[k] ← VictimPortLoc(VictimSwitch[j])
        if VictimPort[k] is connected to a host then
            AttackerIP ← host.IP
        end if
    end if
end for
```

(C) Deployment of defense strategies

In summary, the attack host can be located based on Algorithm 2. In order to effectively mitigate the LDDoS attack flow,We send the defense rules to the SDN controller through the REST API interface, and the controller then sends the flow table according to the rules to block malicious traffic from the attacking host, and the defense rules are shown in Table 1. This method can effectively prevent the attacker from injecting the attack traffic into the network again after tampering with the IP address.

5.3. Time complexity analysis of LR-STGCN defense framework

The time complexity of LDDoS defense framework is mainly composed of the following key components: wavelet transform algorithm, GCN-GRU algorithm, host source localization algorithm, and sliding window algorithm. The time complexity of the wavelet transform algorithm for feature selection is $O(n)$, where n is the number of samples

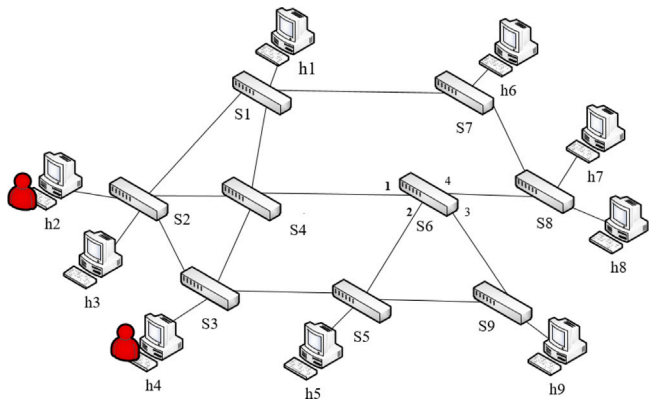


Fig. 9. Network topology.

in the detection window. The time complexity of GCN-GRU detection model is composed of GCN and GRU, and the time complexity of GCN layer is $O(m \cdot f \cdot f' + e \cdot f)$, where m is the number of nodes, e is the number of edges, f is the feature dimension of each node. f' is the transformed feature dimension. The time complexity of the GRU layer is $O(m \cdot H \cdot (D+H))$, where D is the input dimension and H is the hidden dimension of the GRU. Assuming that the number of GCN layers is l, the time complexity of GCN-GRU is $O(l \cdot (m \cdot f \cdot f' + e \cdot f)) + O(m \cdot H \cdot (D+H))$. The time complexity of the host-source location algorithm is $O(j)$, where j is the number of victim switches. The time complexity of the double sliding window algorithm is $O(k \cdot n)$, where k is the number of ports of victim switches. Then the overall time complexity of the whole framework is $O(n) + O\left(l \cdot (m \cdot f \cdot f' + e \cdot f)\right) + O(m \cdot H \cdot (D+H)) + O(j) + O(k \cdot n)$. This reflects the comprehensive computational cost in the process of LDDoS attack detection and mitigation, and provides an effective and scalable solution for LDDoS attack defense in large-scale SDN networks.

6. Experiments and results

6.1. Environment setup

This paper is based on a Linux system and uses Mininet simulation software (Mininet, 2024), RYU controller (Anon, 2024a), and Open vSwitch switches (Anon, 2024b) to build an experimental environment. The experimental topology is shown in Fig. 9, where hosts h2 and h4 are attack hosts, and the rest of the hosts are normal hosts. The Open vSwitch switches S4 and S6, S6 and S8 are bottleneck links, with a link bandwidth of 10 Mb/s, and the rest of the links have a bandwidth of 100 Mb/s. The latency of all links is 10 ms.

h1 and h5 establish normal TCP connections with h9 and h6, respectively, and h3 establishes a normal UDP connection with h8. h1, h3, and h5 are the senders, and h6, h8, and h9 are the receivers, and the UDP and TCP traffic is generated by the D-ITG tool (Anon, 2024c), and the packet rate of sending TCP traffic for each normal host is 500 packets/s, and the packet rate of sending UDP traffic is 150 packets/s, and the packet size is 512 Byte. The attacking host sends UDP attack traffic to h7 using the LDDoS attack program written in Python, and the attack parameters are shown in Table 2. To collect effective LDDoS data, this paper uses the Wireshark tool to collect data at each switch port. A sliding window technique is utilized for data capture, with a time span of 1s. The detection window DW is set to 10 s, and the sampling interval Δt is configured at 100 ms. The experiment lasts for 600 s and yields 6000 samples (including normal flow and LDDoS attack flow) are collected. We selected 80% of the data as the training set and the remaining 20% as the test set. There are no attacks in the network except for the LDDoS attack.

Table 2
Parameters of the attack.

Pulse rate R (Mb/s)	Pulse length L/ms	Attack period T/s
10	300	1

6.2. Evaluation metrics

To verify the effectiveness of the GCN-GRU model classifier, this article selects evaluation indicators such as Accuracy, Precision, Recall, F1-score, etc. For performance testing, such as Eqs. (29)–(35). Among them, TP represents the true case, which refers to the number of positive cases classified by the model, which is the number of positive cases; FP represents false positive, which refers to the number of positive cases classified by the model but negative cases; TN represents true negative examples, which refers to the number of negative examples classified by the model; FN represents false negative examples, which refers to the number of positive examples that the model classifies as negative. Treat attack flow as a positive example and normal flow as a negative example.

$$Precision = \frac{TP}{TP + FP} \times 100\% \tag{29}$$

$$Recall = \frac{TP}{TP + FN} \times 100\% \tag{30}$$

$$Acccuracy = \frac{TP + TN}{TP + TN + FP + FN} \times 100\% \tag{31}$$

$$F_1 - score = \frac{2 \times Precision \times Recall}{Precision + Recall} \times 100\% \tag{32}$$

Detection Rate I_{DR}, False Alarm Rate I_{FAR}, and Missing Alarm Rate I_{MAR} are commonly used indicators to evaluate intrusion detection results. To effectively evaluate the performance of LDDoS attack detection, we use the following three metrics:

(1) Detection Rate I_{DR} (DR): the value of correctly predicting the number of attack flows divided by the actual total number of attack flows. The larger the value, the better the detection effect of LDDoS attacks.

$$I_{DR} = TP/(TP + FN) \tag{33}$$

(2) False Alarm Rate I_{FAR} (FAR): Refers to the number of normal flows predicted as attack flows and the ratio to the actual number of normal flows. The higher the value, the higher the prediction rate of normal flow.

$$I_{FAR} = FP/(FP + TN) \tag{34}$$

(3) Missing Alarm Rate I_{MAR} (MAR): the value of dividing the number of incorrectly predicted attack flows by the actual total number of attack flows. The higher the value, the better the detection effect of the LDDoS attack.

$$I_{MAR} = FN/(TP + FN) \tag{35}$$

6.3. Attack detection experiment

6.3.1. Experimental setup

The deep learning framework we have chosen is PyTorch 1.13.1, and the programming language is Python 3.8. The experimental environment is a laptop with Intel Core i7-10510U CPU and 16 GB RAM. We use the Adam optimizer to optimize the model parameters, setting the learning rate to 0.001 and epoch to 200. The proposed anomaly detection model was established using Pytorch.

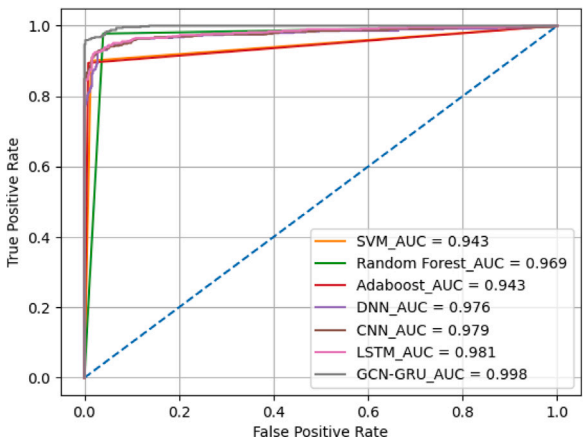


Fig. 10. ROC of GCN-GRU and different detection methods.

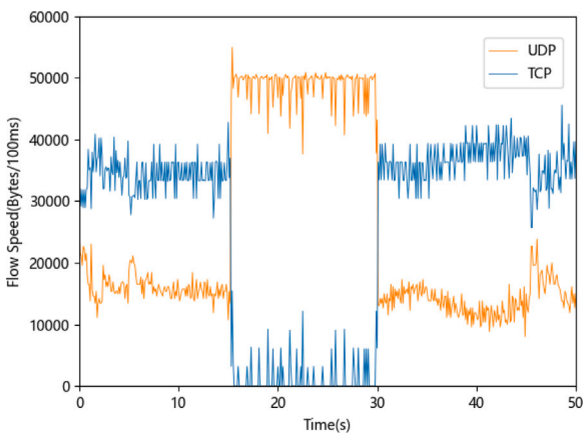


Fig. 11. Trend of traffic changes at port1 of Switch S6.

6.3.2. Method comparison

To verify the robustness of the proposed detection model GCN-GRU, this paper uses the dataset we collected and the InSDN dataset (Elsayed et al., 2020) for verification. The InSDN dataset simulates real internal and external SDN attack scenarios, covering recent common attack types, Such as DoS, DDoS, Brute Force Attack, Malware, Probe, Exploitation, and Web attack. It provides valuable data resources for intrusion detection research in SDN environments, aiding researchers in developing and evaluating various security algorithms and systems. In this experiment, the GCN-GRU model is compared with Support Vector Machine (SVM), Random Forest (RF), Adaptive Boosting (AdaBoost), Deep neural Network (DNN), Convolutional Neural network (CNN), and Long Short-Term Memory network (LSTM). The evaluation metrics of different algorithms are shown in Table 3. Experimental results show that the GCN-GRU model achieves the best accuracy, precision, recall, and F1-score on both our dataset and the InSDN dataset, indicating that the GCN-GRU model exhibits strong robustness.

6.3.3. ROC and AUC

Next, we plotted the ROC curve. The abscissa of the curve is the False Positive Rate (FPR) and the ordinate is the True Positive Rate (TPR). According to the ROC curve, we can see the effect of the model at any threshold. The ROC curve divides the whole plot into two parts. The area under the curve is called AUC (Area Under the Curve) and indicates the accuracy of the detection. The higher the AUC value (the larger the area under the curve), the better the prediction accuracy. The closer the curve is to the top-left corner, the better the model generalizes. As shown in Fig. 10, our method performs well on the ROC curve and all the AUC values are higher than the other methods. Specifically, the AUC value of the GCN-GRU detection method reaches 0.998, which is close to 1. It shows that the GCN-GRU detection method has great competitiveness and can effectively improve the ability of network traffic anomaly detection.

Table 3
Evaluation metrics of different algorithms.

Dataset	Method	Accuracy	Precision	Recall	F1-score
Our Dataset	SVM	94.39%	96.45%	90.06%	93.15%
	RF	95.99%	95.94%	95.94%	95.94%
	Adaboost	94.39%	97.10%	89.45%	93.12%
	DNN	94.09%	96.37%	91.48%	93.86%
	CNN	95.80%	97.08%	94.32%	95.68%
	LSTM	94.29%	95.42%	92.90%	94.14%
	GCN-GRU	**97.25**%	**97.30** %	**97.27**%	**97.25**%
InSDN	SVM	95.66%	96.48%	96.16%	96.32%
	RF	98.86%	98.65%	99.43%	99.04%
	Adaboost	95.79%	97.24%	95.59%	96.41%
	DNN	94.73%	96.21%	94.80%	95.50%
	CNN	96.99%	98.61%	96.27%	97.43%
	LSTM	96.33%	96.73%	97.06%	96.90%
	GCN-GRU	**99.53**%	**99.49** %	**99.57**%	**99.53**%

Table 4
Results of ablation experiment.

Method	Accuracy	Precision	Recall	F1-score
Model w/o GCN	95.10%	96.25%	93.71%	94.96%
Model w/o GRU	96.87%	96.92%	96.89%	96.87%
Model w/o GCN w/CNN	95.90%	96.89%	94.73%	95.79%
GCN-GRU	**97.25**%	**97.30**%	**97.27**%	**97.25**%

Table 5
LDDoS attack detection results.

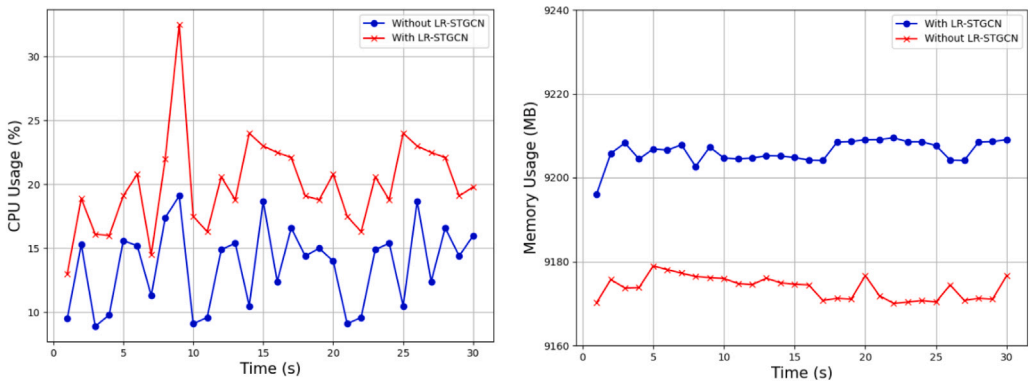
Method	DR	FAR	MAR
ADMS (Tang et al., 2022a)	94.23%	8.76%	5.77%
P&F (Tang et al., 2022b)	94.93%	2.98%	5.07%
Joint Feature (Zh et al., 2017)	89.96%	5.37%	10.04%
MF-Adaboost (Tang et al., 2020b)	95.94%	5.77%	4.06%
Our method	**97.30**%	**0.91**%	**2.70**%

6.3.4. Ablation experiment

To verify the effectiveness of this method, we conducted model ablation experiments. Specifically, we sequentially remove one model component from the model and compare it with the complete GCN-GRU model. We will name these models as follows:

- Model w/o GCN: This model does not have a GCN component.
- Model w/o GRU: This model does not have a GRU component.
- Model w/o GCN, w/CNN: This model does not have GCN components and consists of CNN and GRU components.

The results of ablation experiments are shown in Table 4. Experimental results show that removing either component from the model results in significant performance degradation. This verifies the advantages of the GCN-GRU model in spatio-temporal information fusion, which can effectively capture the dependencies between network nodes and the temporal characteristics of attack traffic. Specifically, the GCN component is responsible for extracting the spatial dependencies in the network graph structure, while the GRU component focuses on learning the temporal dynamics of the traffic data. The combination of the two makes the model perform well in detecting Low-rate Distributed attacks (LDDoS). Compared with single GCN, GRU, CNN, and other deep learning methods, the GCN-GRU model shows obvious advantages in detection effect and can identify LDDoS attack behaviors more accurately.



(a) CPU Usage **(b)** Memory Usage

Fig. 12. Deployment cost of our scheme.

Table 6
Performance comparison of different LDDoS attack defense frameworks.

Framework	Recall	FAR	Response	CPU	Memory
ADMS (Tang et al., 2022a)	92.15%	8.76%	AVE = 14.5 s	4.6–12.1%	22–36 MB
P&F (Tang et al., 2022b)	96.17%	2.98%	AVE = 9.39 s	8.7–10.7%	24–30 MB
SoftGuard (Xie et al., 2019)	95.74%	5.09%	AVE = 18 s	3.8–13.7%	28–36 MB
Our method	**97.27**%	**0.91**%	AVE = 11.2 s	3.2–13.5%	26–39 MB

6.3.5. Comparison with different LDDoS attack detection methods

To verify the effectiveness of our method, we compared it with the previous LDDoS attack detection method,such as ADMS (Tang et al., 2022a), P&F (Tang et al., 2022b), Joint Feature (Zh et al., 2017) and MF-Adaboost (Tang et al., 2020b). We compared our proposed method with the state-of-the-art methods in terms of Detection Rate (DR), False Alarm Rate (FAR), and False Alarm Rate (MAR). The comparison results are shown in Table 5. The experimental results show that our proposed GCN-GRU model has the highest score on all evaluation indicators, proving that the performance of this detection method is superior to other LDDoS attack detection methods. Therefore, GCN-GRU can accurately detect LDDoS attacks.

6.4. Defense effect evaluation

In order to effectively alleviate the LDDoS attack, we add a mitigation module to the RYU controller to implement the detection and defense mechanism mentioned above. This article uses Algorithm 2 (where t' is 60 s, τ is 50 ms, N_0 is 6, λ is 5, and K is 6)and Algorithm 3 to determine the attack hosts h2 and h4. Finally, the attack paths h2 → S2 → S4 → S6 → S8 → h7 and h4 → S3 → S4 → S6 → S8 → h7 are found.

To verify the effectiveness of the method, we start the mitigation module in the RYU controller and use the Wireshark tool to observe the network flow rate changes at port 1 of S6. As can be seen in Fig. 11, the UDP network traffic at port 1 of switch S6 rises sharply, and the TCP traffic drops sharply at about 15 s. After about 15 s (the detection time set in this paper is 15 s), the LR-STGCN system detects the attack and defends it by issuing the flow rule command. The network traffic returns to the initial state, indicating that the mitigation is effective.

6.5. Performance comparison

We conducted comparative experiments of LR-STGCN with ADMS (Tang et al., 2022a), P&F (Tang et al., 2022b), and SoftGuard (Xie et al., 2019), using an experimental environment similar to ours. The specific comparison results are shown in Table 6. In these four groups of experiments, our method demonstrates exceptional efficiency, achieving a Recall of 97.27% and a FAR as low as 0.91%. This makes it particularly suitable for applications requiring high detection accuracy and low false positives. In terms of real-time performance, the average response time of our method is 11.2 s (from the attack trigger to the end of the attack), which is slightly higher than the 9.39 s of the P&F method. However, compared to ADMS and SoftGuard, our method shows more stability in network latency, with overall moderate performance. Moreover, as shown in Fig. 12, our method performs well in terms of CPU utilization, only 3.2%–13.5%, which is much lower than the other three methods, making it a significant advantage in long-running or resource-constrained scenarios. In terms of memory footprint, the memory usage of our method is between 26MB and 39MB, which is comparable to other methods, and all belong to lightweight systems. In summary, our framework demonstrates strong performance and holds significant application value, especially for LDDoS attack detection and mitigation in SDN environments.

7. Conclusion

In this paper, we propose a framework for detecting and mitigating LDDoS attacks. First, the LR-STGCN framework selects features by comprehensively considering both attack performance and attack traffic characteristics. Then, the GCN-GRU model is used to detect attack traffic from both spatial and temporal perspectives. Compared with other detection methods, this approach not only improves detection accuracy and reduces false positives but also assists in tracing LDDoS attacks. Finally, the double sliding window method and the switch's energy E_a are utilized for attack tracing, and the OpenFlow protocol is employed to achieve real-time mitigation of LDoS attacks. Experimental results demonstrate that the proposed method achieves high detection accuracy and low false positive rates, enabling accurate and real-time mitigation of LDDoS attacks. In future work, we will focus on improving the adaptive capability of the detection model to enable it to flexibly address various emerging LDDoS attack patterns. Meanwhile, we plan to thoroughly investigate the scalability and distributed detection performance of the LR-STGCN framework in real multi-controller SDN environments, aiming to tackle the complex challenges in large-scale network scenarios.

CRediT authorship contribution statement

Jin Wang: Writing – review & editing, Writing – original draft, Validation, Conceptualization. **Liping Wang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Jin Wang received the master's degree from the School of Computer Science, Zhongyuan University of Technology in 2018. She is currently pursing a Ph.D. with the College of Computer Science and Technology, Zhejiang University of Technology. Her research interests include Software-Defined Networking, Intrusion detection and machine learning.

Liping Wang received the Ph.D. degree in computer science and technology from Zhejiang University, China, in 2005. She is currently a Professor with the College of Computer Science and Technology, Zhejiang University of Technology. Her research interests include network security and machine learning.